



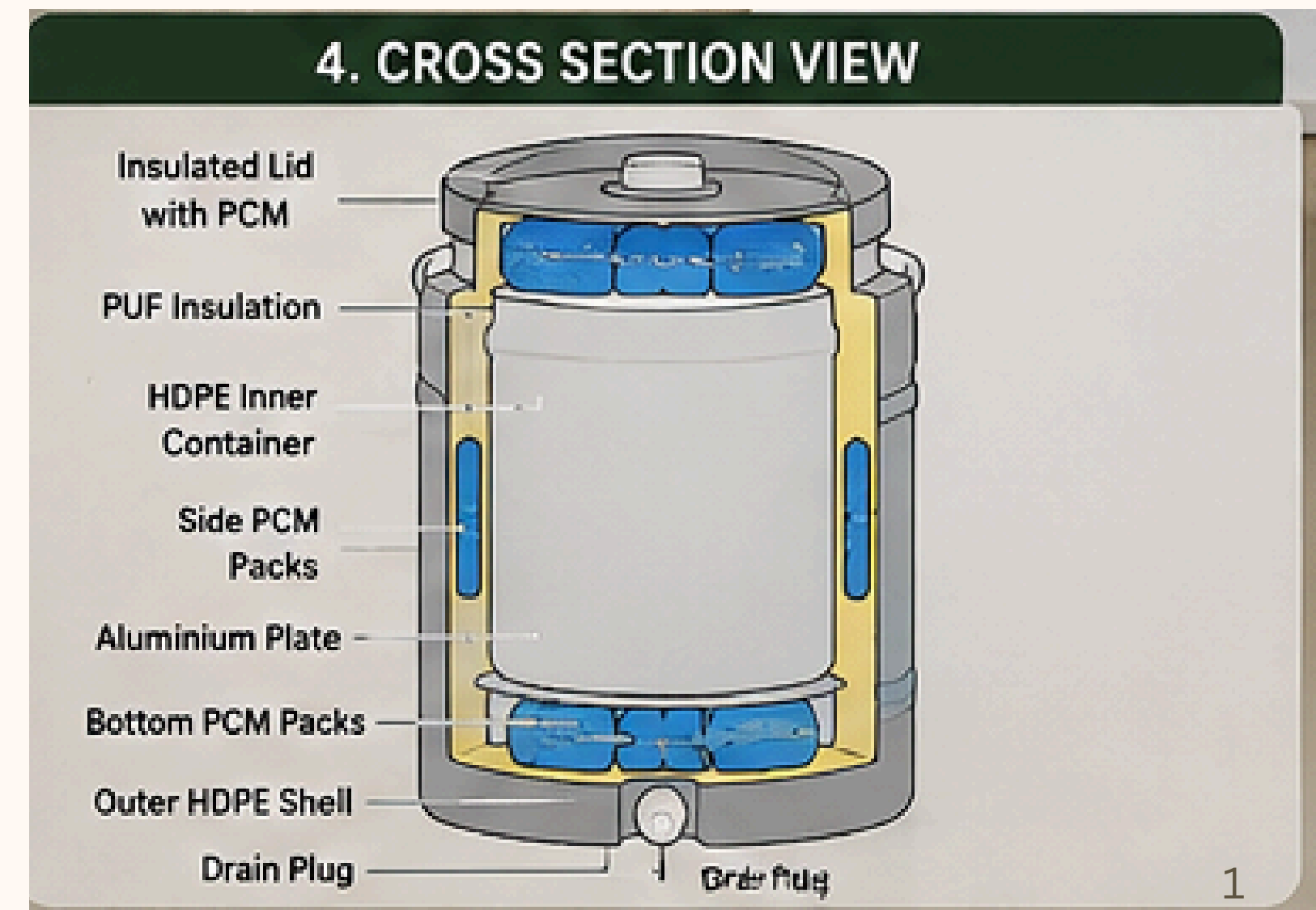
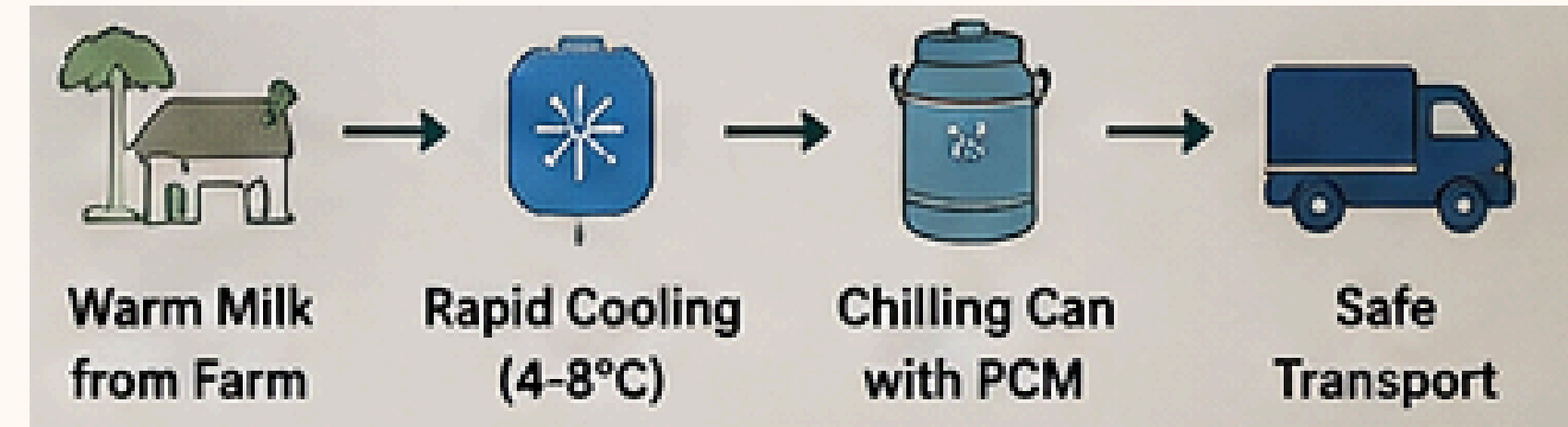
KILOWATT
KREW

SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

- Problem Statement ID – 26110
- Problem Statement Title- Development of a Low cost Light weight Milk Chilling Can for Small Scale Dairy Farmers
- Theme- Agriculture, FoodTech and Rural Development
- PS Category- Hardware
- Team ID- SIH26-A0H-T192
- Team Name (Registered on portal) Kilowatt Krew





KILOWATT
KREW

PROBLEMS IN HAND

MILK QUALITY:Temperature rise can accelerate microbial growth and deterioration of milk quality.Inadequate temperature control between collection and chilling can reduce the quality of milk reaching the processing facility.

Mechanical:Rough roads and hilly terrain expose milk cans to vibration and shocks.The current existing can cant withstand the pressure and which build heat inside the can

Thermal: Maintaining milk at the required 4–8°C during transportation is challenging.Milk is exposed to ambient heat during collection and transit.Long transportation durations increase the cooling requirement.

Refrigerant Impact: Conventional bulk chillers rely extensively on chemical refrigerants like Freon or Ammonia, which carry high global warming potential and fail to offer an eco-friendly option for smaller operations.

PROPOSED SOLUTION DAIRY CHILL

LOW-COST LIGHTWEIGHT MILK CHILLING CAN For Safe Milk Transportation in Rural Areas



Why We Stand Out

We don't just transport milk — we actively protect its quality throughout transportation.”

- **Active + Passive Cooling** — cooling system brings the temperature down, while insulation and PCM help maintain it.
- **Temperature Monitoring** — enables continuous awareness of milk temperature.
- **Transport Protection** — shock-absorbing construction improves durability on rough roads.
- **Designed for Rural & Hilly Transport** — focuses on portability, ruggedness and practical field conditions.



SMART INDIA
HACKATHON
2026

KEY FEATURES

- **Active Cooling System** — rapidly brings milk toward the required 4–8°C range.
- **PCM-Based Thermal Retention** — helps maintain temperature and reduces dependence on continuous active cooling.
- **Real-Time Temperature Monitoring** — continuously monitors the milk temperature.
- **Shock-Resistant Construction** — designed to withstand vibration and rough transportation conditions.
- **Energy-Conscious Operation** — insulation + PCM reduce the cooling load on the active system.
- **Multi-Layer Thermal Insulation** — minimizes heat gain from the surroundings.



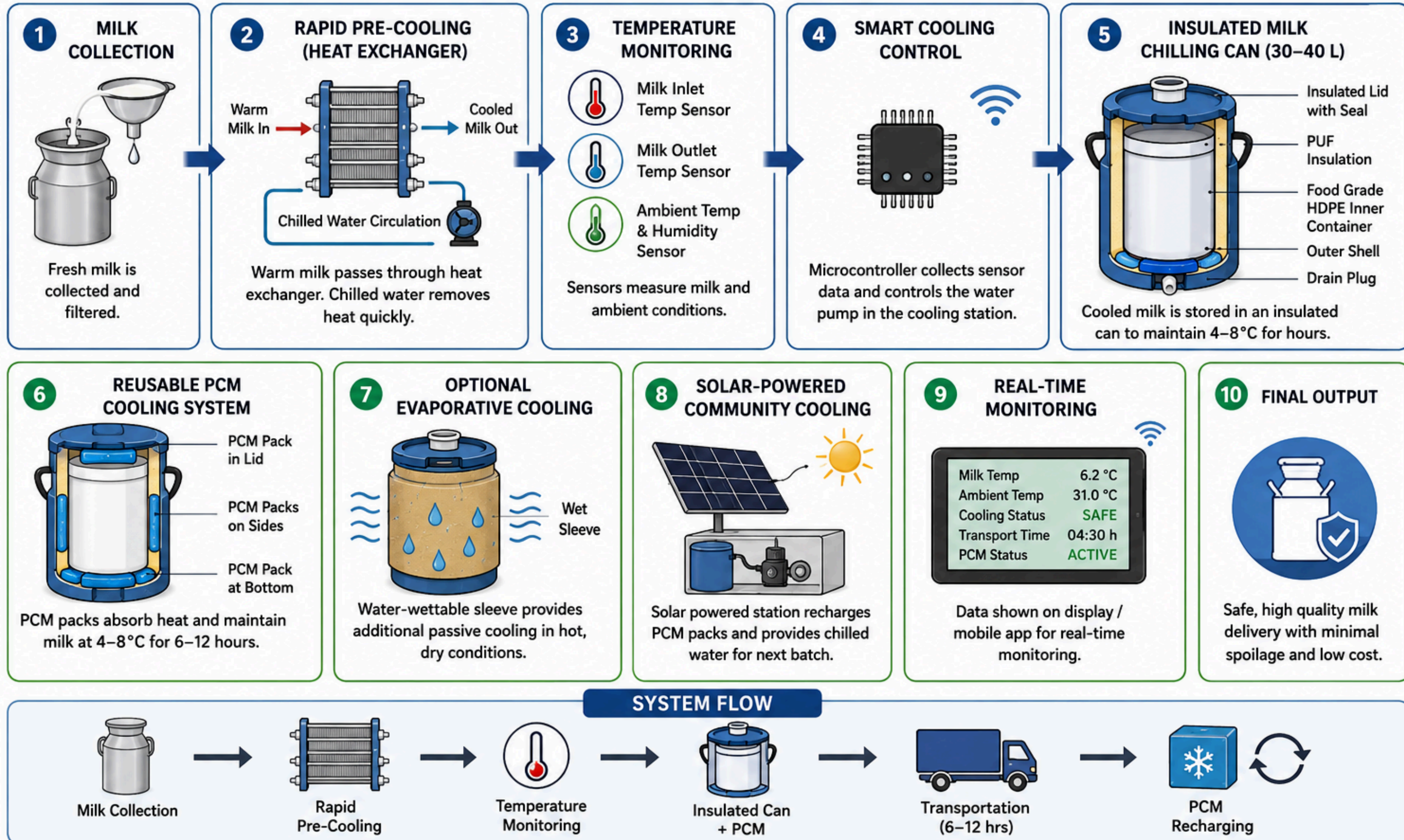
KILOWATT
KREW

TECHNICAL APPROACH

Low-Cost Passive Milk Chilling & Transportation System



SMART INDIA
HACKATHON
2026





KILOWATT
KREW

FEASIBILITY AND VIABILITY



SMART INDIA
HACKATHON
2026

Technical Feasibility

- Uses readily available materials: HDPE, PUF insulation, aluminium, PCM/ice packs.
- Rapid pre-cooling can be achieved using a heat exchanger and chilled-water circulation
- Modular design makes the system easy to repair and maintain.

Operational Feasibility

- Simple to load, transport and clean.
- PCM packs can be recharged at village collection centres.
- Solar-powered community cooling stations can support remote areas.

Economic Viability

- Lower expected cost than conventional powered milk chilling systems.
- Reusable PCM packs reduce recurring cooling costs.
- Shared cooling/recharging stations distribute infrastructure cost among multiple farmers.

Social Viability

- Suitable for rural, hilly and remote dairy supply chains.

Environmental Viability

- Minimal electricity use during transportation.
- Solar-powered recharge can further reduce energy requirements.

ECONOMIC FEASIBILITY

- Low material cost and simple manufacturing process.
- Affordable for small and marginal farmers.
- Reduces milk spoilage and quality loss → saves money.



FINANCIAL VIABILITY

- Low production cost enables good profit margin.
- Affordable selling price ensures higher adoption.
- Recurring revenue through replacement cooling packs and accessories.



Conclusion: The proposed milk chilling can is both FEASIBLE to develop and VIABLE for real-world implementation, ensuring improved milk quality, reduced losses and higher income for dairy farmers.



KILOWATT
KREW



SMART INDIA
HACKATHON
2026

IMPACT AND BENEFITS

REDUCE MILK SPOILAGE

- Rapid pre-cooling followed by insulated transportation helps to maintain milk at 4-5 C
- Reduces bacterial growth, quality degradation and milk wastage

LIGHTWEIGHT & AFFORDABLE

- Lightweight HDPE/ composite construction reduce handling and transportation effort
- Designed to be significantly more economical than conventional insulated chilling container



3. ENERGY FREE OPERATION

- No electricity required during transport.
- Works on passive cooling using ice/PCM and insulation.

HIGHER FARMER INCOME

- Less milk rejection due to poor quality
- Better milk quality can improve the value received by small and marginal dairy farmers



1. IMPROVED MILK QUALITY

Maintains milk at safe temperature (4–8°C), reducing bacterial growth and spoilage. Results in better taste, higher shelf life and premium pricing.



2. REDUCED MILK SPOILAGE AND LOSSES

Minimizes quality degradation during collection and transportation. Reduces quantity and financial losses for farmers.

SUSTAINABLE & REUSEABLE

- Reusable cooling cartridges reduce dependence on disposable ice
- A shared solar-power/community cooling station can recharge the cooling modules for repeated use

ELECTRICITY-INDEPENDENT TRANSPORTATION

- Reusable PCM/ice cartridges maintain the temperature without continuous power during transportation
- particularly useful in remote hilly and North-Eastern regions with unreliable electricity

BETTER RURAL MILK SUPPLY CHAIN

Enables safe transportation from Farm to village collection centre to dairy facility

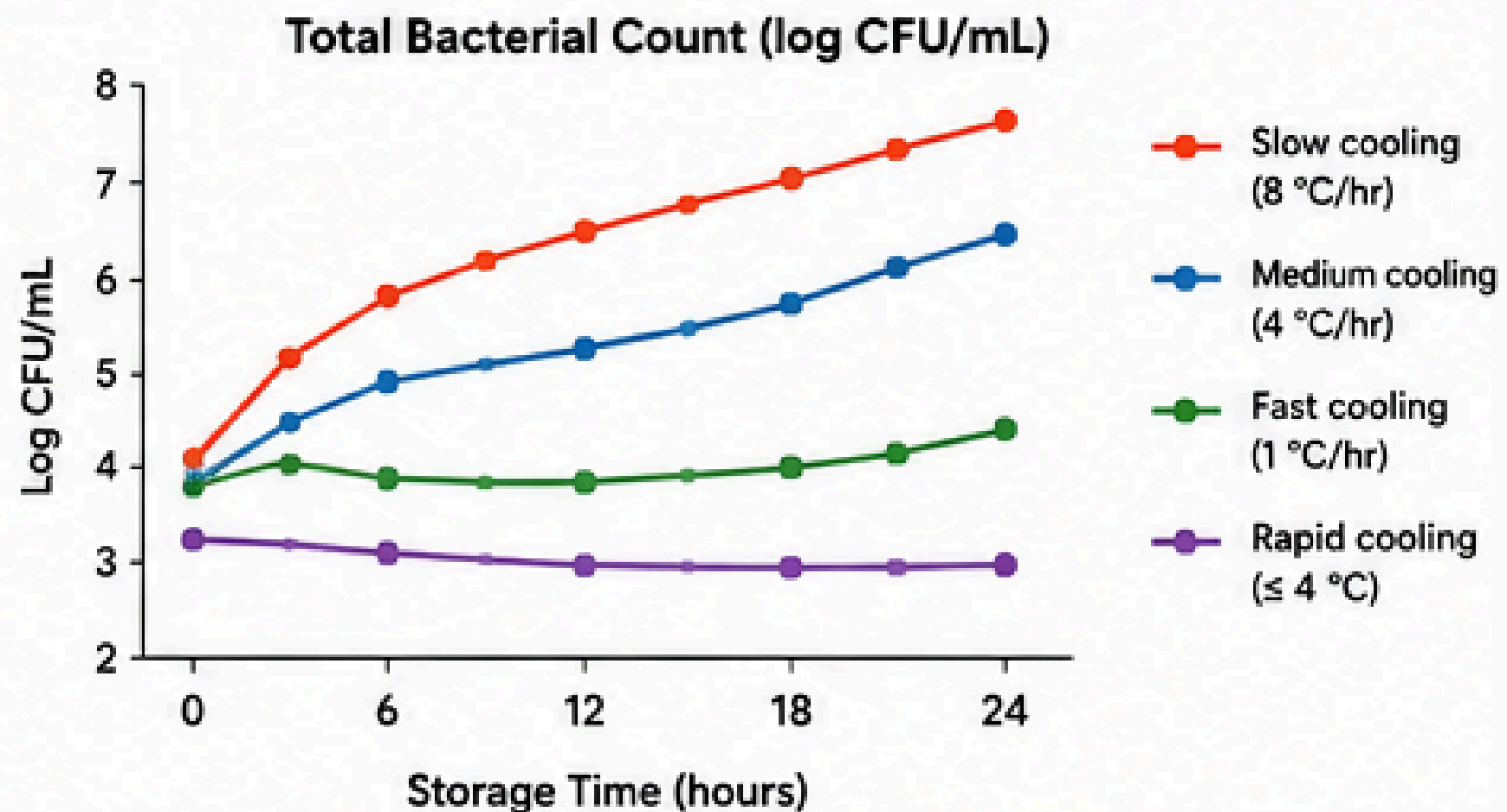
- Reduces dependence on expensive bulk milk cooling infrastructure at every farm



4. EASY TO CLEAN AND MAINTAIN

- Smooth food-grade inner surface.
- Leak-proof design ensures hygiene and easy washing.

Paludetti, L. F. et al. (2018). The effect of different precooling rates and cold storage on milk microbiological quality and composition. Journal of Dairy Science.



3.Ndyabawe, K. & Kisaalita, W. S. (2014). Diffusion of an evaporative cooler innovation among smallholder dairy farmers of Western Uganda. Technology in Society.

Chaomuang, N. et al. (2025). Thermal performance evaluation of water-based gel packs and insulating covers for raw milk preservation. Applied Food Research

Thermal Performance Results

Cooling Setup	Initial Milk Temp (°C)	Milk Temp After 8 h (°C)	Temp Range Maintained
No Cooling (Control)	30.2	18.6	Poor
Ice Only	30.1	6.2	Moderate
Gel Pack Only	30.0	5.4	Good
Gel Pack + Insulating Cover	30.1	4.3	Excellent (4–8 °C)



Implication for Our Project:

Validates the use of water-based gel/PCM packs along with insulation to maintain milk within the safe temperature range for longer duration.